



◀ LOCAL TIDES

Tidal range is influenced by local geography as well as by the alignment of the Sun and Moon. Some coasts have huge tidal ranges that are caused by tidal water being forced into funnel-shaped bays and estuaries, and then draining away again as the tide ebbs. The record is held by the Bay of Fundy in eastern Canada, seen here at low tide, which experiences a maximum tidal range of 50 ft (16 m).

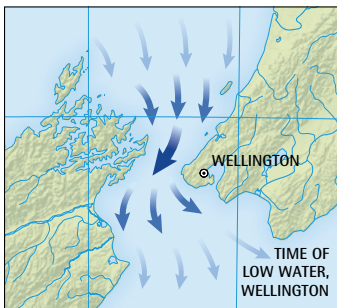


▲ LOWS AND HIGHS

As the tide floods into the Bay of Fundy in eastern Canada—shown at low tide at the top of the page—it can rise at a rate of up to 13 ft (4 m) an hour until the tide peaks just six hours later, as seen here. By contrast, tide levels on many other coasts barely rise at all, and in parts of semi-enclosed seas such as the Mediterranean, the difference between low and high tide is barely perceptible.

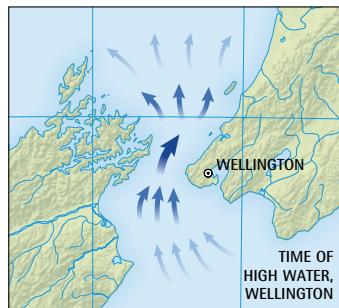


TIDAL STREAMS



FLOOD TIDE

As the tide rises, water flows along coasts and floods into bays and river estuaries. These tidal streams can flow fast around headlands and through narrow straits, such as the Cook Strait between New Zealand's North and South Island. Such local tidal streams are often more powerful than ocean currents.



EBB TIDE

When the tide level starts to fall, water drains out of bays and estuaries in the ebb tide, and the direction of tidal streams along the coast reverses. In order to make their voyages possible, sailors have to know when these reversals occur each day, so they can take advantage of favorable tides.

RACES AND WHIRLPOLS ►

Strong tidal streams pouring through narrow channels and around headlands can cause tidal races—fast, turbulent flows marked by whirlpools and steep, breaking stopper waves that stay in place as water flows through them. This whirlpool is a spin-off from the Saltstraumen tidal race on the northwestern coast of Norway, one of the most powerful tidal races in the world.

